NAME : N.S.MUNITEJA

REG NO :192312388

EX.NO:1: SIMULATION OF FDMA IN MATLAB DATE

Objective 1:

Code:

```
clc;
clear;
close all;

fs = 1000;
t = 0:1/fs:1;

TBW = 600;
Users = 3;
CBW = TBW/Users;

fc = [100 300 500];

m = sin(2*pi*10*t);

s1 = m.*cos(2*pi*fc(1)*t);
s2 = m.*cos(2*pi*fc(2)*t);
s3 = m.*cos(2*pi*fc(3)*t);

subplot(2,1,1)
plot(t,s1,'r',t,s2,'g',t,s3,'b')
grid on
title('FDMA Channel Allocation')
```

xlabel('Time (s)')

ylabel('Amplitude')

subplot(2,1,2)

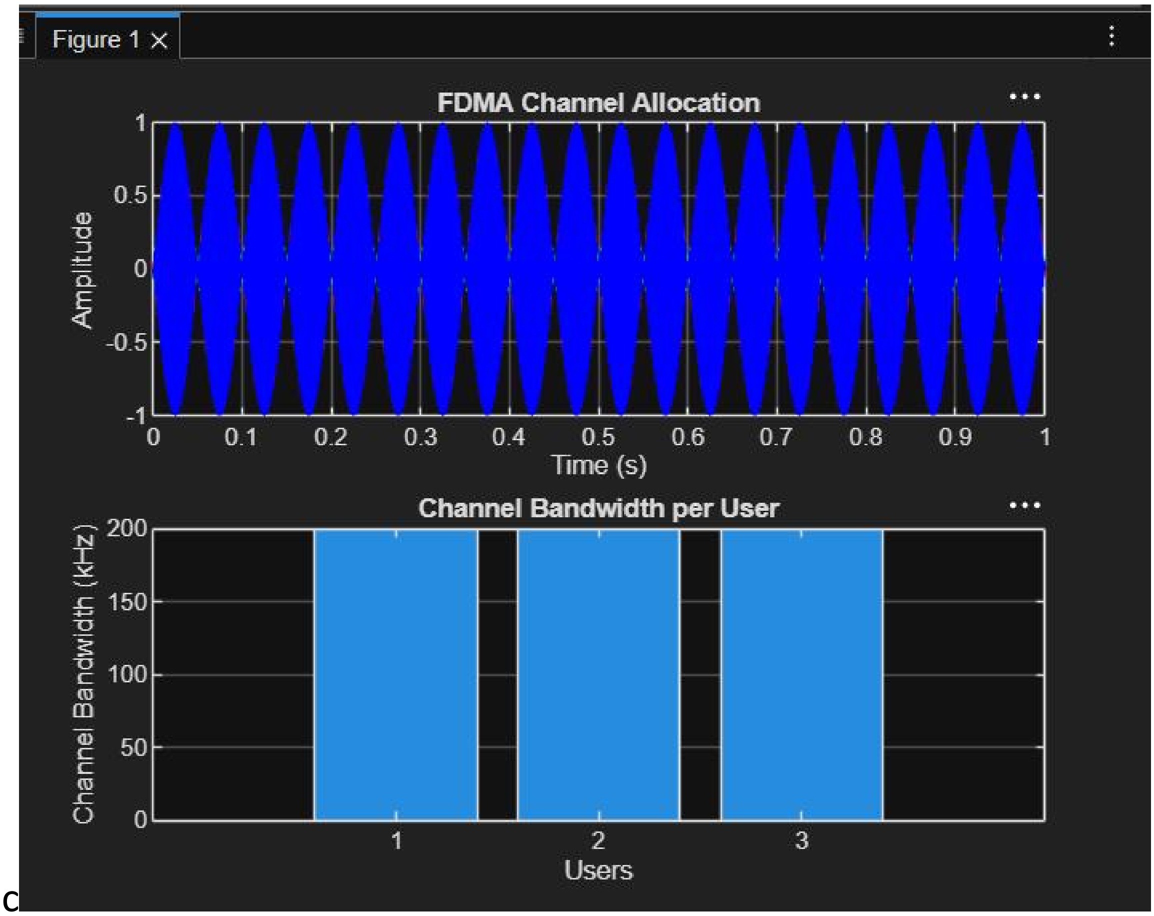
bar(1:Users, CBW*ones(1,Users))

grid on

xlabel('Users')

ylabel('Channel Bandwidth (kHz)')

title('Channel Bandwidth per User')



c

Objective 2:

Code:

clc;

clear;

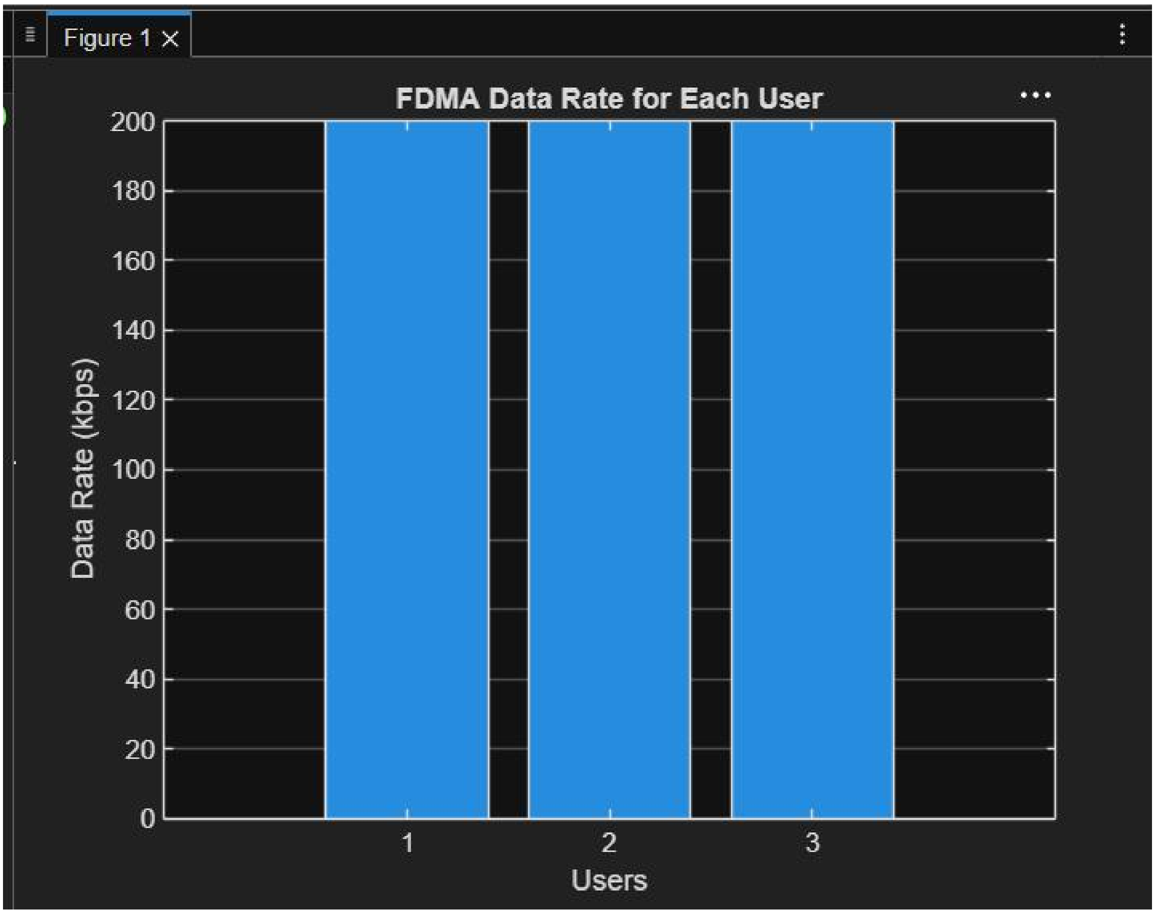
```
close all;

BW = 200;         % Channel Bandwidth (kHz)
Users = 3;
M = 2;            % BPSK Modulation

DataRate = BW * log2(M);    % Data Rate (kbps)
Rate = DataRate * ones(1,Users);

disp('User Data Rate (kbps)')
disp(table((1:Users)', Rate', ...
    'VariableNames', {'User','DataRate_kbps'}))

figure
bar(1:Users, Rate)
xlabel('Users')
ylabel('Data Rate (kbps)')
title('FDMA Data Rate for Each User')
grid on
```



Objective 3:

Code:

```
clc; clear; close all;

Ps = 1; % Signal Power (W)

Pn = [0.1 0.05 0.01]; % Noise Power (W)

Users = 1:3;

SNR = 10*log10(Ps./Pn); % SNR in dB

disp('User Noise Power(W) SNR(dB)')

disp([Users' Pn' SNR'])

figure

bar(Users,SNR)

xlabel('Users')

ylabel('SNR (dB)')

title('Signal-to-Noise Ratio of FDMA Users')
```

grid on

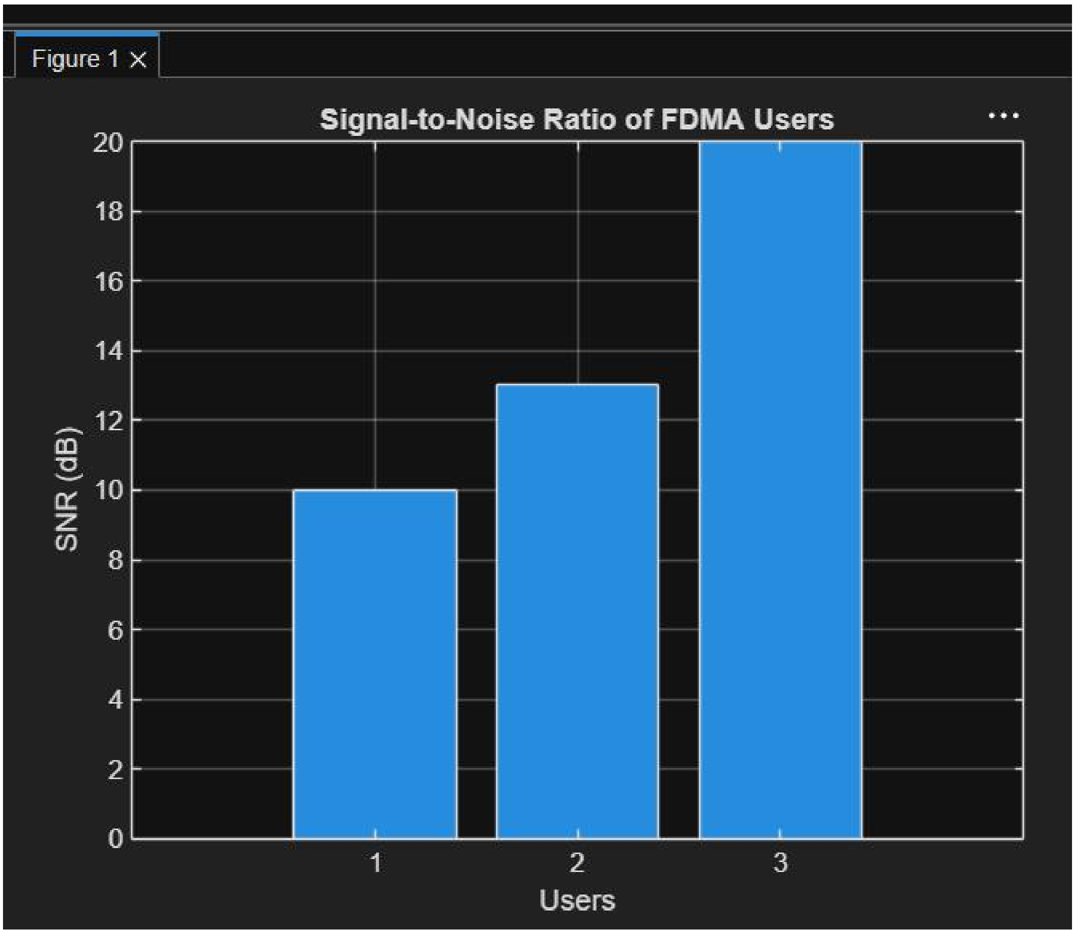


Figure 1
Signal-to-Noise Ratio of FDMA Users
SNR (dB)
20
18
16
14
12
10
8
6
4
2
0
1
2
3
Users